

Bruno Fernandes Garcia

PhD Student in Space Geophysics | Solar Physics | INPE, Brazil

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Google Scholar: scholar.google.com/citations?user=1WRcLkkAAAAJ |

Lattes: lattes.cnpq.br/7725799765396213

Research Interests

Solar active regions - Non-Linear Force-Free Field (NLFFF) and Magnetohydrostatic (MHS) extrapolation - Physics-Informed Neural Networks (PINNs) - Magnetic topology - Space Weather - Heliophysics

Education

PhD in Space Geophysics

2024 – Present

Instituto Nacional de Pesquisas Espaciais (INPE), São José dos Campos, Brazil

- Supervisor: Dr. Joaquim Eduardo Rezende Costa (Director, EMBRACE Program)
- Co-supervisor: Dr. Luis Eduardo Antunes Vieira (Head, GSST Project)
- Sandwich Doctorate: Queen Mary University of London, UK (Supervisor: Dr. Enrico Camporeale)
- Funding: CAPES scholarship
- Keywords: Solar Physics, Solar Corona, PINNs, Artificial Intelligence
- Award: SAGE III Featured Paper Award (2025)

Professional Master's in Physics Teaching

2020 – 2022

Universidade Federal de Alfenas (UNIFAL-MG), Alfenas, Brazil

- Thesis: Solar Physics as a Convergence of Physics Topics in Secondary Education
- Supervisor: Dr. Samuel Bueno Soltau

Bachelor's in Physics (Teaching Degree)

2016 – 2019

Universidade Federal de Lavras (UFLA), Lavras, Brazil

- CAPES scholarship (PIBID — Teaching Initiation Program, 1152h)

Technician in Mechatronics

2010 – 2012

CEFET-MG, Varginha, Brazil

Research Experience

PhD Researcher — Solar Physics

Mar 2024 – Present

INPE, São José dos Campos, Brazil

- NLFFF magnetic field extrapolation of solar active regions using Physics-Informed Neural Networks (PINNs) trained with the PyMHS library (PyTorch)
- Magnetic field line tracing (RK4 integrator), identification of null points, separatrices, and helicity distributions across multiple ARs and time steps
- Focus region: NOAA AR3663 and other complex active regions

- Member of the GSST (Galileo Space Solar Telescope) team: solar instrumentation project at INPE
- Student representative of the Space Geophysics Postgraduate Programme
- Institutional environment spanning EMBRACE's operational space weather mission and GSST observational infrastructure

Scientific Initiation Fellow (PIBIC)

2012 – 2013

Instituto de Aeronáutica e Espaço (IAE/CTA), Brazil

- Project: Evaluation of thermophysical properties of CRFC composites for aerospace use (504h)

Technical Skills

- Programming: Python (PyTorch, NumPy, SciPy, Matplotlib, Astropy), Bash
- Methods: Physics-Informed Neural Networks, NLFFF extrapolation, RK4 field line tracing, GPU computing
- Data: SDO/HMI magnetograms, solar EUV imaging data, space weather indices
- Tools: Git, Jupyter, Linux/HPC environments

Publications

- Garcia, B.F.; Soltau, S.B. (2021). Física solar: uma experiência de ensino remoto durante a pandemia. Research, Society and Development, 10, e49910313495.

Web of Science: 4 citations | Google Scholar: 5 citations

Awards & Recognition

- SAGE III Featured Paper Award — III Semana Acadêmica da Geofísica Espacial, INPE (2025)

Languages

- Portuguese — Native
- English — Functional (reading, writing, speaking, listening)

Previous Professional Experience

Physics Teacher — State Schools of Minas Gerais

2019 – 2023

Multiple public schools, MG, Brazil (Alfenas, São Sebastião do Paraíso, Três Corações, Varginha)

- Full-time secondary school physics teacher in the Minas Gerais state school system
- Physics preparatory course instructor (ESPCEX 2022, Colégio Padrão, Três Corações)